Data treatment and Transformation

Business Analytics, 1e

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Learning Objectives (LOs)

LO 1 Apply data preparation techniques to inspect data quality, handle missing values or impute the data as necessary.

LO 2 Transformation of data and Reducing number of variables (categorical versus numeric)

Data Analytic process

1 Ask

- Ask effective questions
- Define the problem
- Use structured thinking
- Communicate with others

2 Prepare

- Understand how data is generated and collected
- Identify and use different data formats, types, and structures
- Make sure data is unbiased and credible
- Organize and protect data

3 Process

- Create and transform data
- Maintain data integrity
- Test data
- Clean data
- Verify and report on cleaning results

4 Analyze

- Use tools to format and transform data
- Sort and filter data
- Identify patterns and draw conclusions
- Make predictions and recommendations
- Make data-driven decisions

5 Share

- Understand visualization
- Create effective visuals
- Bring data to life
- Use data storytelling
- Communicate to help others understand results

6 Act

- Apply your insights
- Solve problems
- Make decisions
- Create something new

Introductory Case: Retail Customer Data

- Catherine is a marketing manager at Organic Food Superstore.
- She would like to use the company's marketing dollars to market a new line of Asian-inspired meals to college-educated millennials.
- She has acquired a representative sample.

CustID	Gender	Race	BirthDate	. . .	Channel
1530016	Female	Black	12/16/1986	. . .	SM
1531136	Male	White	5/9/1993	. . .	TV
⋮	⋮	⋮	⋮	⋮	⋮
1579979	Male	White	7/5/1999	. . .	Referral

- Catherine will use the data for the following tasks.
  1. Identify Organic Food Superstore's college-educated millennial customers.
  2. Compare profiles of female and male college-educated millennial customers.

Data Management

- Data management or data wrangling is the process of retrieving, cleansing, integrating, transforming, and enriching data to support subsequent analysis.
  - Transform raw data into a format that is more appropriate and easier to analyze
- Objectives:
  - improving data quality,
  - Reducing time/effort required to perform analytics,
  - Reveal the true intelligence in the data

Why to learn Data Management?

- Analytics professionals no longer rely solely on the corporate IT department for data retrieval and preparation.
- 50% to 80% of time an analytics professional spent is to collect and prepare unruly data before analytics can be applied.
- They have to become more self-reliant and possess the necessary skills for data wrangling and analysis, broader skill sets than just statistics.
- This allows organizations to address business problems much more quickly.
- In Data Science, we refer to this process as EDA.

EDA Process

- Exploratory Data Analysis: involves looking at and visualizing data to understand its main features, find patterns, and discover how different parts of the data are connected.
- usually done before starting more detailed statistical analysis or building models
- we can divide EDA into three types:
  - Univariate: Focusing on one variable to understand its characteristics, e.g. histogram or bar chart, summary stats.
  - Bivariate: understand how two variables interact, e.g. **scatter/Line plots**, **correlation/Covariance**, and contingency table.
  - Multivariate: understand how variables interact with one another, which is crucial for most statistical modeling techniques, e.g. **pair plots**, **Principal Component Analysis (PCA)**, and Spatial map.

Steps of EDA

Step 0: Understand the Problem and the Data

Step 1: Inspect the Data

- **Identify data types** for each variable (like numerical, categorical, etc.),
- **Check for missing values**
- **Look for errors** or inconsistencies, such as invalid values, mismatched units, or outliers/anomalies.

Step 2: Handle Missing Data and errors

- Decide whether to remove missing data (listwise deletion) or impute (fill in) the missing values.
- Use appropriate imputation methods like mean/median or regression, or machine learning techniques like KNN or decision trees.

Step 3: Explore Data Characteristics and Summary Stats.

- For numerical data, you should check on the distribution, calculate summary statistics like mean, median, mode, standard deviation, skewness, and kurtosis for identify any irregular patterns or issues.

Step 4: Perform Data Transformation

- Common transformation techniques include Scaling or normalizing (e.g., min-max scaling or standardization).
- Encoding categorical variables for machine learning (e.g., one-hot encoding or label encoding).
- Applying mathematical transformations (e.g., logarithmic or square root) to correct skewness or non-linearity.
- Creating new variables from existing ones (e.g., calculating ratios or combining variables).
- Aggregating or grouping data based on specific variables or conditions

Step 5: Visualize Data Relationship (Business Intelligence)

- Help to uncover relationships between variables and identify patterns or trends that may not be obvious from summary statistics alone.

Handing missing values

- Missing values are a common quality problem found in data sets both large and small, e.g.
  - Respondents decline to provide the information due to its sensitive nature
  - Can also be caused by human errors, sloppy data collection, or equipment failures
- Missing values can lead to a significant reduction in the number of usable observations and lead to bias results

Inspecting data

- Once the raw data are extracted from the source (*database, data warehouse, or data lake*), review and inspect the dataset to assess data quality.
- In addition to visually reviewing data, counting and sorting are among the very first tasks most data analysts perform to gain insights into the data.
- Well inspected data enable us to make data preparation:
  - Handling missing values
  - Grouping/Subsetting data

Dealing with missing values

- Basically, there are two strategies for dealing with missing values:
  - omission
  - imputation.
- Omission: complete-case deletion
  - Exclude observations with missing values
  - Appropriate when the amount of missing value is small or concentrated in a small number of observations
- Imputation: replace missing values with some reasonable values

Data preparation: Data Imputation

- For numerical values, replace missing values with the mean/average value across relevant observations.
  - Easy to implement
  - Doesn't increase the variability in the data set
  - If a large number of values are missing, mean imputation will likely distort the relationships among variables, leading to biased results
    - Apply advance technique, e.g. machine learning, or use business logic.
- For categorical variables
  - May choose Random, Mode, kNN or Predictive (ML-technique)
  - Create an "Unknown" or "Missing" category
  - If the variable that has many missing values is deemed unimportant, the variable may be excluded from the analysis.
    - If missing is low (<5%): Mode or "Missing" category may be applied
    - If missing is medium (<20%) use kNN, otherwise (>20%) using Predictive.

Example for missing data handling

EXAMPLE 2.2

Sarah Johnson, the manager of a local restaurant, has conducted a survey to gauge customers' perception about the eatery. Each customer rated the restaurant on its ambience, cleanliness, service, and food using a scale of 1 (lowest) to 7 (highest). Table 2.4 displays a portion of the survey data.

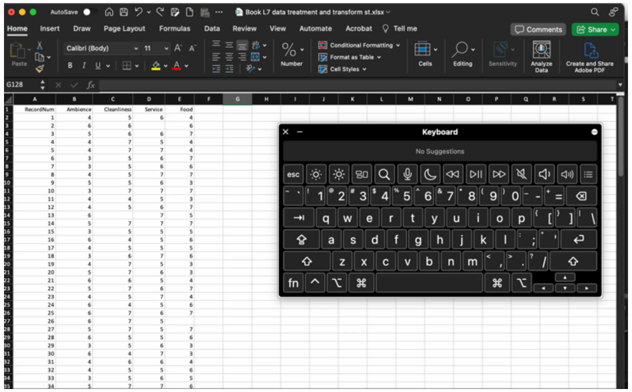
TABLE 2.4 Restaurant Reviews

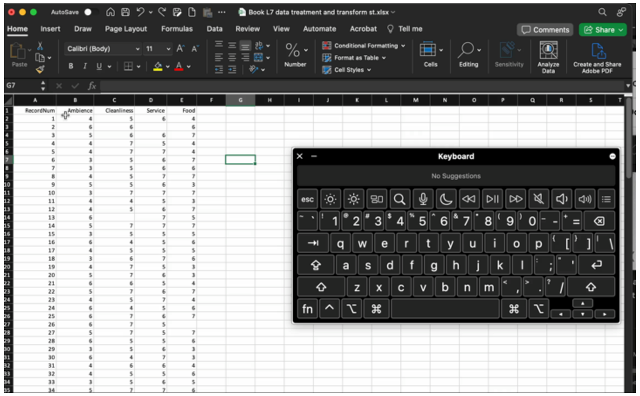
RecordNum	Ambience	Cleanliness	Service	Food
1	4	5	6	4
2	6	6		6
⋮	⋮	⋮	⋮	⋮
150	3	5	6	7

Sarah notices that there are a number of missing values in the survey. Use the ***Restaurant_Reviews*** data to first detect the missing values. Then use both omission and imputation strategies to handle the missing values.

Data Preparation in Excel

- Omission (deletion) using Excel
  – Using Home>Conditional Formatting>New Rule to format Blank cells with filled color.
  – Using =countblank(), to count number of missing vlaues in a column
  – Manual select rows and delete or Using filters to show only Blank rows for deleting
- Imputing
  – Using “=average()” for columns with missing values
  – Using Find&Replace in menu “Home/Find&Select/Replace” per each column to replace blank with corresponding mean value.





Data Subsetting

The process of extracting portions of a data set that are relevant to the analysis

- For time series data, we may choose to create subsets of recent observations and observations from the distant past in order to analyze them separately.
  - E.g. Stock data of 25 years, getting only the latest 3 years for analysis.
- Subsetting also be used for eliminating unwanted data such as observations that contain missing values, low quality data, or outliers.
- Sometimes, subsetting involves excluding variables out of the observations.
  - Irrelevant to the problem
  - Contain redundant information
  - Excessive amounts of missing values

Recap: Data Preparation

- Example: Revisit the introductory case.

In the introductory case, Catherine Hill wants to gain a better understanding of Organic Food Superstore's customers who are college-educated millennials, born on or after 1/1/1982 and before 1/1/2000. She feels that sex, household size, income, total spending in 2021, total number of orders in the past 24 months, and channel through which the customer was acquired are useful for her to create a profile of these customers. Use Excel to first identify college-educated millenial customers in the ***Customers*** data file. Then, create subsets of female and male college-educated millenial customers. The synopsis that follows this example provides a summary of the results.

shape (49, 6) ....

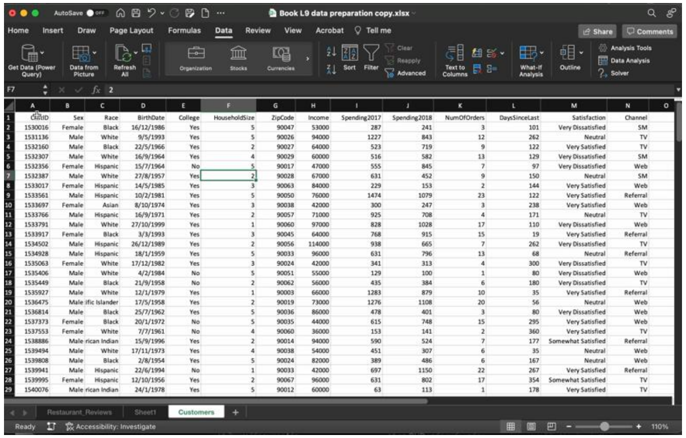
	Gender	HouseholdSize	Income	Spending2021	NumOfOrders	Channel
1	Male	5	94000	843	12	TV
10	Male	1	97000	1028	17	Web
12	Male	2	114000	665	7	TV
15	Male	5	55000	100	1	Web
22	Male	2	94000	524	7	Referral

shape (25, 6) ....

	Gender	HouseholdSize	Income	Spending2021	NumOfOrders	Channel
0	Female	5	53000	241	3	SM
6	Female	3	84000	153	2	Web
11	Female	3	64000	915	15	Referral
14	Female	3	42000	313	4	TV
50	Female	5	97000	911	16	Web

Excel’s solution

- Using filter to select Customers with College degree (yes), Birthdate after 31/12/1981 and before 1/1/2000 (Use customer filter Date filter > between)
- Copy and paste the filtered data to a new worksheet as a subsetting result.
- Delete unrelevant data according to the thought of Catherine.
- Next, use Sort&Filter to sort Gender column, then copy and paste data of each gender to new worksheets as a result for subsetting Male and Female data.



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LO 1 Apply data preparation techniques to inspect data quality, handle missing values or impute the data as necessary.

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Transforming Categorical Data

- We use labels or names to identify the distinguishing characteristics of a categorical variable.
  - Categorical variables can also be defined by two or more categories, e.g. Gender → Male, Female, genderfree, …
  - We use nominal and ordinal measurement scales to represent categorical variables
- While categorical variables are known to represent less sophisticated levels of measurement, they are often the most important variables in the analysis.
- As many analysis techniques are limited in their abilities to handle categorical data directly, steps to simplify or transform categorical data into numerical formats are often performed prior to analysis.

Transforming Categorical Data: Reduction

- Sometimes, nominal or ordinal variables come with too many categories presents a number of potential problems:
  - If a variable has some categories that rarely occur, it is difficult to capture the impact of these categories accurately.
  - Variables with too many categories pull down model performance.
- An effective strategy is category reduction, where we collapse some of the categories to create fewer nonoverlapping categories.
- Determining the appropriate number of categories often depends on the data, context, and disciplinary norms
  - Categories with a similar impact may be combined.
  - Categories with very few observations may be combined to create the “Other” category. A mass in “Other” category may help reveal patterns and relationships in data.

EXAMPLE 2.6

After gaining some insights from the ***Customers*** data set, Catherine would like to analyze race. However, in its current form, the data set would limit her ability to perform a meaningful analysis given the large number of categories of the race variable; plus some categories have very few observations. As a result, she needs to perform a series of data transformations to prepare the data for subsequent analysis. Use Excel to create a new category called Other that represents the two least-frequent categories.

SOLUTION:

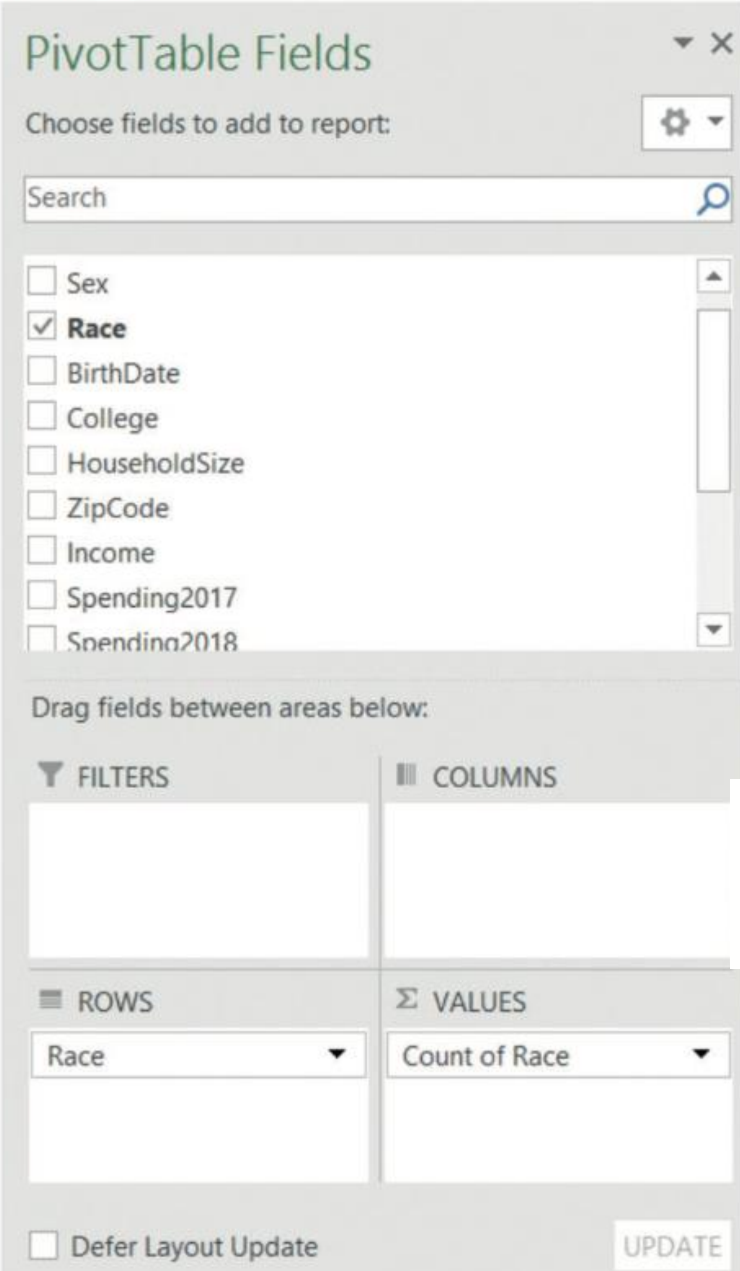
Using Excel

a. Open the ***Customers*** data file.

b. We use a pivot table to display the frequency for each race category. A pivot table allows us to summarize raw data in a more manageable way. Select cell P1 and choose **Insert > PivotTable**. In the *Create PivotTable* dialog box, we choose the *Select a table or range* option and specify the *Table/Range* value to be Customers!$A$1:$N$201, which refers to the entire ***Customers*** worksheet. We then select the *Existing Worksheet* option and specify the *Location* value to be Customers!$P$1. This means that the pivot table will start in cell P1 of the ***Customers*** worksheet. Click **OK**. The *PivotTable Fields* pane appears, as shown in Figure 2.9. First, drag the Race field to the *ROWS* box, and then drag the Race field to the *VALUES* box. The *ROWS* box lists the race categories, whereas the *VALUES* box lists the frequency of each race category.

c. We now combine the two least-frequent race categories into the Other category using the **IF** function. In the ***Customers*** worksheet, enter the column heading NewRace in cell O1. In cell O2, enter the formula =IF(OR(C2="American Indian", C2="Pacific Islander"), "Other", C2). The **OR** function returns TRUE if the results of any of the logical tests are TRUE. The formula states that if the race category in cell C2 is either American Indian or Pacific Islander, then change the race to Other; otherwise, record the current race category in cell C2. Fill the range O3:O201 with the formula in O2. Verify that the 19th record represents the first customer in the Other category.

FIGURE 2.9 PivotTable Fields pane



Microsoft Corporation

FIGURE 2.10 Pivot table for race

Row Labels	Count of Race
American Indian	5
Asian	15
Black	57
Hispanic	41
Pacific Islander	3
White	79
Grand Total	**200**

We now combine the two least-frequent race categories into the Other category using the **IF** function. In the ***Customers*** worksheet, enter the column heading NewRace in cell O1. In cell O2, enter the formula =IF(OR(C2="American Indian", C2="Pacific Islander"), "Other", C2).

Transforming Categorical to Numeric data

- In many analytical models, categorical variables must first be converted into numerical variables.
  - For some models, dealing with numerical data help to avoid the complexities of the semantics pertaining to each category.
- Dummy encoding (for linear regression or stats model)
  - creating **k-1** binary variables for a categorical feature with **k** unique categories.
  - It assumes a value of 1 for one of the categories (called reference or benchmark category)
- One-hot encoding
  - Creates **k** binary variables for **k** unique categories.
  - No category is dropped.
  - Used in **machine learning models** like decision trees, neural networks, and clustering.

Sample Data of Red, Blue, Green

Color	Dummy_1 (Blue)	Dummy_2 (Green)
Red	0	0
Blue	1	0
Green	0	1

Color	Red	Blue	Green
Red	1	0	0
Blue	0	1	0
Green	0	0	1

EXAMPLE 2.7

For the new Asian-inspired meal kits, Catherine feels that understanding the channels through which customers were acquired is important to predict customers' future behaviors. In order to include the Channel variable in her predictive model, Catherine needs to convert the Channel categories into dummy variables. Because web banner ads are probably the most common marketing tools used by Organic Food Superstore, she plans to use the Web channel as the reference category and assess the effects of other channels in relation to the Web channel. Use Excel to create the relevant dummy variables for the Channel variable.

SOLUTION:

Using Excel

a. Open the ***Customers*** data file.

b. Using the Web channel as the reference category, we create three dummy variables for the Channel variable because there are four categories. Enter the variable names Channel_Referral, Channel_SM, and Channel_TV in cells O1, P1, and Q1, respectively.

c. Enter the formula =IF(N2="Referral", 1, 0) in cell O2. Fill the range O3:O201 with the formula in O2. Enter the formula =IF(N2="SM", 1, 0) in cell P2. Fill the range P3:P201 with the formula in P2. Enter the formula =IF(N2="TV", 1, 0) in cell Q2. Fill the range Q3:Q201 with the formula in Q2. Verify that the dummy variable values for the first observation are 0, 1, and 0, respectively.

Transforming Categorical Data: Scoring

- To create category scores.
  - most appropriate *if the data are ordinal* and have natural, ordered categories.
- Usually we assume equal increments between the category scores
- Example: In customer satisfaction surveys, we often use ordinal scales such as very dissatisfied, somewhat dissatisfied, neutral, somewhat satisfied, and very satisfied to indicate the level of satisfaction.
  - we can recode the categories using numbers 1 - 5 with 1 being very dissatisfied and 5 being very satisfied.

EXAMPLE 2.8

For the new Asian-inspired meal kits, Catherine wants to pay attention to customer satisfaction. As the customer satisfaction ratings represent ordinal data, she wants to convert them to category scores ranging from 1 (Very Dissatisfied) to 5 (Very Satisfied) to make the variable more readily usable in predictive models. Use Excel and R to create category scores for the Satisfaction variable.

SOLUTION:

Using Excel

a. Open the ***Customers*** data file.

b. Enter the column heading Satisfaction_Score in cell O1. Enter the formula =IF(M2="Very Satisfied", 5, IF(M2="Somewhat Satisfied", 4, IF(M2="Neutral", 3, IF(M2="Somewhat Dissatisfied", 2, 1)))) in cell O2. Fill the range O3:O201 with the formula in O2. The scores are now ordered based on the degree to which the customer is satisfied with Organic Food Superstore's service. See Example 2.4 for further information on the nested IF statement. Verify that the first four satisfaction scores are 1, 3, 5, and 1, respectively.

Transforming Numerical Data

- It is performed to meet the requirements of statistical and data mining techniques used for the analysis.
- Sometimes, it makes sense to group a vast range of numerical values into a small number of "**bins**."

Transforming Numerical Data: Binning

Binning is the process of transforming numerical variables into categorical variables (grouping into a small number of groups or bins)

- bins should be consecutive and nonoverlapping so that each numerical value falls into only one bin.
- Binning can be an effective way to reduce noise in the data if we believe that all observations in the same bin tend to behave the same way.
- Outliers in the data will be part of the last bin and, and therefore, will not distort subsequent data analysis.
- We can create bins of equal counts (frequency), where individual bins have the same number of observations.
- Bins are often created to have equal intervals.

- **Data :** 0, 4, 12, 16, 16, 18, 24, 26, 28
- **Equal width**
  - Bin 1: 0, 4 [-,10)
  - Bin 2: 12, 16, 16, 18 [10,20)
  - Bin 3: 24, 26, 28 [20,+)
- **Equal frequency**
  - Bin 1: 0, 4, 12 [-, 14)
  - Bin 2: 16, 16, 18 [14, 21)
  - Bin 3: 24, 26, 28 [21,+)

EXAMPLE 2.4

In order to better understand her customers, Catherine Hill would like to perform the RFM analysis, a popular marketing technique used to identify high-value customers. RFM stands for recency, frequency, and monetary. The RFM ratings can be created from the DaysSinceLast (recency), NumOfOrders (frequency), and Spending2021 (monetary) variables.

Following the 80/20 business rule (i.e., 80% of your business comes from 20% of your best customers), for each of the three RFM variables, Catherine would like to bin customers into five equal-size groups, with 20% of the customers included in each group. Each group is also assigned a score from 1 to 5, with 5 being the highest. Customers with the RFM rating of 555 are considered the most valuable customers to the company.

In addition to the RFM binning, Catherine would like to bin the Income variable into five equal intervals. Finally, she would like to start a tiered membership system where different services and rewards are offered to customers depending on how much they spent in 2021. She would like to assign the bronze membership status to customers who spent less than $250, silver membership status to those who spent $250 or more but less than $1,000, and the gold membership status to those who spent $1,000 or more.

Use Excel to bin variables according to Catherine's specifications. Summarize the results.

Transforming Numerical Data

- Example: Binning result

Column S	Column T
Spending	Membership
0	Bronze
250	Silver
1000	Gold

CustID	. . .	RFM	Binned Income	Membership Tier
1530016	. . .	411	1	Bronze
1531136	. . .	244	3	Silver
⋮	⋮	⋮	⋮	⋮
1579979	. . .	434	3	Silver

Conclusion

- A data wrangling and management are discussed for enabling a well prepared data for analysis.
- Treatment of missing values and subsetting are introduced.
- Data transformation for categorical and numerical data are also discussed.